

Advertisement

Time limit: 1.00 s

Memory limit: 512 MB

A fence consists of n vertical boards. The width of each board is 1 and their heights may vary. You want to attach a rectangular advertisement to the fence. What is the maximum area of such an advertisement?

Input

The first input line contains an integer n : the width of the fence.

After this, there are n integers $k_1, k_2, \dots, k_n$: the height of each board.

Output

Print one integer: the maximum area of an advertisement.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq k_i \leq 10^9$

Example

Input	Output
8 4 1 5 3 3 2 4 1	10